

(*This notebook computes and bounds the infinite
Euler product of $\frac{A}{p}$'s in the $U^{\{r\}}$ -sum.*)

```
In[1]:= SetDirectory[NotebookDirectory[]];
<< gln.m
```

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⋯ **SetDelayed:** Tag UpperTriangularMatrixQ in UpperTriangularMatrixQ[m_] is Protected.

GL(n)PACK loaded.

```
In[3]:= Schu[n_, d_, nota_ : a] :=
SchurPolynomial[Table[nota[i], {i, 1, d}], Join[ConstantArray[0, d - 2], {n}]]
```

```
In[4]:= Symma3 = (a[1] - a[2]) (a[1] - a[3]) (a[2] - a[3]);
```

```
In[5]:= Symb3 = (b[1] - b[2]) (b[1] - b[3]) (b[2] - b[3]);
```

```
In[6]:= (*U= psi_{1}(p)/p^{u_{1}}, V= psi_{2}(p)/p^{u_{2}}, Z= 1/p^z*)
```

```
In[7]:= FF2[d_, g_] := Sum[Schu[M + g, d, b] * V^M, {M, 0, Infinity}]
```

```
In[8]:= FF1[d_, g_] := Sum[Schu[M + g, d, a] * U^M, {M, 0, Infinity}]
```

```
In[9]:= FF3[d_, g_] := Schu[g, d, a] * FF2[d, g] + Schu[g, d, b] * FF1[d, g] - Schu[g, d, a] * Schu[g, d, b]
```

```
In[10]:= InvL[X_, d_, nota_ : a] := Product[(1 - nota[i] X), {i, 1, d}]
```

```
In[11]:= InvRSL[X_, d_, nota1_ : a, nota2_ : b] :=
Product[Product[(1 - nota1[i] * nota2[j] X), {i, 1, d}], {j, 1, d}]
```

```
In[12]:= NormaSum[d_] :=
Cancel[Sum[FF3[d, g] * Z^g, {g, 0, Infinity}] * InvL[U, d, a] * InvL[V, d, b] * InvRSL[Z, d, a, b]]
```

```
In[13]:= (*PGL(2)*)
```

```
In[14]:= NormaSum[2]
```

```
Out[14]=
```

$$\begin{aligned}
& 1 - U V a[1] \times b[1] - U V a[2] \times b[1] + U^2 V a[1] \times a[2] \times b[1] - U Z a[1] \times a[2] \times b[1] + U V Z a[1] \times a[2] b[1]^2 - \\
& U V a[1] \times b[2] - U V a[2] \times b[2] + U^2 V a[1] \times a[2] \times b[2] - U Z a[1] \times a[2] \times b[2] + U V^2 a[1] \times b[1] \times b[2] - \\
& V Z a[1] \times b[1] \times b[2] + U V Z a[1]^2 b[1] \times b[2] + U V^2 a[2] \times b[1] \times b[2] - V Z a[2] \times b[1] \times b[2] - \\
& U^2 V^2 a[1] \times a[2] \times b[1] \times b[2] + 4 U V Z a[1] \times a[2] \times b[1] \times b[2] - Z^2 a[1] \times a[2] \times b[1] \times b[2] - \\
& U^2 V Z a[1]^2 a[2] \times b[1] \times b[2] + U Z^2 a[1]^2 a[2] \times b[1] \times b[2] + U V Z a[2]^2 b[1] \times b[2] - U^2 V Z a[1] a[2]^2 b[1] \times b[2] + \\
& U Z^2 a[1] a[2]^2 b[1] \times b[2] - U V^2 Z a[1] \times a[2] b[1]^2 b[2] + V Z^2 a[1] \times a[2] b[1]^2 b[2] - U V Z^2 a[1]^2 a[2] b[1]^2 b[2] - \\
& U V Z^2 a[1] a[2]^2 b[1]^2 b[2] + U V Z a[1] \times a[2] b[2]^2 - U V^2 Z a[1] \times a[2] \times b[1] b[2]^2 + V Z^2 a[1] \times a[2] \times b[1] b[2]^2 - \\
& U V Z^2 a[1]^2 a[2] \times b[1] b[2]^2 - U V Z^2 a[1] a[2]^2 b[1] b[2]^2 + U^2 V^2 Z^2 a[1]^2 a[2]^2 b[1]^2 b[2]^2
\end{aligned}$$

```
In[15]:= (*The following computes the arithmetic factor \frac{A}{p} of the paper*)
```

```

In[16]:= GL2inSym = (SymmetricReduction[#, {a[1], a[2]}, {A[1], ω}][[1]] & /@
  (SymmetricReduction[#, {b[1], b[2]}, {B[1], ω}][[1]] & /@ NormaSum[2])) /. ω ^ k_ . :> ω

Out[16]=
1 - U^2 V^2 ω - Z^2 ω + U^2 V^2 Z^2 ω + U V^2 ω A[1] - V Z ω A[1] - U^2 V Z ω A[1] + U Z^2 ω A[1] + U V Z ω A[1]^2 +
  U^2 V ω B[1] - U Z ω B[1] - U V^2 Z ω B[1] + V Z^2 ω B[1] - U V A[1] × B[1] - U V Z^2 ω A[1] × B[1] + U V Z ω B[1]^2

In[17]:= Collect[GL2inSym, DeleteCases[Times @@ ({A[1], B[1]}^#) & /@ Tuples[Range[0, 2], 2], 1]]

Out[17]=
1 - U^2 V^2 ω - Z^2 ω + U^2 V^2 Z^2 ω + (U V^2 ω - V Z ω - U^2 V Z ω + U Z^2 ω) A[1] + U V Z ω A[1]^2 +
  (U^2 V ω - U Z ω - U V^2 Z ω + V Z^2 ω) B[1] + (-U V - U V Z^2 ω) A[1] × B[1] + U V Z ω B[1]^2

In[18]:= Factor[(U V^2 ω - V Z ω - U^2 V Z ω + U Z^2 ω) A[1] + (U^2 V ω - U Z ω - U V^2 Z ω + V Z^2 ω) B[1]]

Out[18]=
-((U V - Z) ω (-V A[1] + U Z A[1] - U B[1] + V Z B[1]))

In[19]:= (*Unramified bounding*)

In[20]:= UnrListMono2 = List @@ (Expand[GL2inSym /. ω -> 1]);

In[21]:= UnrListexpo2 = Exponent[#, p] & /@ (List @@ UnrListMono2 /.
  {A[1] -> p^Th, B[1] -> p^Th, U -> p^(-1/2), V -> p^(-1/2), Z -> p^(-1)}) /. Th -> 7/64 // N;

In[22]:= Total[UnrListMono2[[#]] & /@ Flatten@Position[#, <-1 & /@ UnrListexpo2, False]]

Out[22]=
1 - U V A[1] × B[1]

In[23]:= Expand[GL2inSym /. ω -> 0]

Out[23]=
1 - U V A[1] × B[1]

In[24]:= (*PGL(3): the calculation is divided into 3 parts*)

In[25]:= Part1Norm =
  Cancel[Sum[Schu[g, 3, a] * Cancel[FF2[3, g] * Symb3 * InvL[V, 3, b]] * Z^g, {g, 0, Infinity}] *
  InvRSL[Z, 3, a, b]];

In[26]:= Length[Part1Norm]

Out[26]=
360

```

```
In[27]:= Part1Symm = Collect[(SymmetricReduction[#, {a[1], a[2], a[3]}, {A[1], P[1],  $\omega$ }]][1] & /@
    (SymmetricReduction[#, {b[1], b[2], b[3]}, {B[1], Q[1],  $\omega$ }]][1] & /@ Cancel[Part1Norm / Symb3]) /.
     $\omega^k \rightarrow \omega$ , DeleteCases[Times @@ ({A[1], B[1], P[1], Q[1]}^#) & /@ Tuples[Range[0, 2], 4], 1]]
```

```
Out[27]= 1 - 2 Z^3  $\omega$  + Z^6  $\omega$  + (V^2 Z  $\omega$  - V^2 Z^4  $\omega$ ) A[1] + V Z^2  $\omega$  A[1]^2 + (V Z^3  $\omega$  - V Z^6  $\omega$ ) B[1] -
    Z^4  $\omega$  A[1]  $\times$  B[1] + V^2 Z^3  $\omega$  B[1]^2 + V Z^4  $\omega$  A[1] B[1]^2 + (-V Z^2  $\omega$  + V Z^5  $\omega$ ) P[1] + Z^3  $\omega$  A[1]  $\times$  P[1] +
    (-V^2 Z^2  $\omega$  - V^2 Z^5  $\omega$  - V Z^3  $\omega$  A[1]) B[1]  $\times$  P[1] + V^2 Z^4  $\omega$  P[1]^2 + (-V^2 Z^3  $\omega$  + V^2 Z^6  $\omega$ ) Q[1] +
    (-V Z - V Z^4  $\omega$ ) A[1]  $\times$  Q[1] + Z^3  $\omega$  B[1]  $\times$  Q[1] - V Z^3  $\omega$  B[1]^2 Q[1] + (-Z^2 + V Z^2 B[1]) P[1]  $\times$  Q[1] + V Z^3  $\omega$  Q[1]^2
```

```
In[28]:= Part2Norm =
    Cancel[Sum[Schu[g, 3, b] * Cancel[FF1[3, g] * Syma3 * InvL[U, 3, a]] * Z^g, {g, 0, Infinity}] *
    InvRSL[Z, 3, a, b]];
```

```
In[29]:= Length[Part2Norm]
```

```
Out[29]= 360
```

```
In[30]:= Part2Symm = Collect[(SymmetricReduction[#, {a[1], a[2], a[3]}, {A[1], P[1],  $\omega$ }]][1] & /@
    (SymmetricReduction[#, {b[1], b[2], b[3]}, {B[1], Q[1],  $\omega$ }]][1] & /@ Cancel[Part2Norm / Syma3]) /.
     $\omega^k \rightarrow \omega$ , DeleteCases[Times @@ ({A[1], B[1], P[1], Q[1]}^#) & /@ Tuples[Range[0, 2], 4], 1]]
```

```
Out[30]= 1 - 2 Z^3  $\omega$  + Z^6  $\omega$  + (U Z^3  $\omega$  - U Z^6  $\omega$ ) A[1] + U^2 Z^3  $\omega$  A[1]^2 + (U^2 Z  $\omega$  - U^2 Z^4  $\omega$ ) B[1] -
    Z^4  $\omega$  A[1]  $\times$  B[1] + U Z^4  $\omega$  A[1]^2 B[1] + U Z^2  $\omega$  B[1]^2 + (-U^2 Z^3  $\omega$  + U^2 Z^6  $\omega$ ) P[1] + Z^3  $\omega$  A[1]  $\times$  P[1] -
    U Z^3  $\omega$  A[1]^2 P[1] + (-U Z - U Z^4  $\omega$ ) B[1]  $\times$  P[1] + U Z^3  $\omega$  P[1]^2 + (-U Z^2  $\omega$  + U Z^5  $\omega$ ) Q[1] +
    (-U^2 Z^2  $\omega$  - U^2 Z^5  $\omega$ ) A[1]  $\times$  Q[1] + (Z^3  $\omega$  - U Z^3  $\omega$  A[1]) B[1]  $\times$  Q[1] + (-Z^2 + U Z^2 A[1]) P[1]  $\times$  Q[1] + U^2 Z^4  $\omega$  Q[1]^2
```

```
In[31]:= Part3Norm = Cancel[Sum[Schu[g, 3, a] * Schu[g, 3, b] * Z^g, {g, 0, Infinity}] * InvRSL[Z, 3, a, b]];
```

```
In[32]:= Length[Part3Norm]
```

```
Out[32]= 33
```

```
In[33]:= Part3Symm = Collect[(SymmetricReduction[#, {a[1], a[2], a[3]}, {A[1], P[1],  $\omega$ }]][1] & /@
    (SymmetricReduction[#, {b[1], b[2], b[3]}, {B[1], Q[1],  $\omega$ }]][1] & /@ Part3Norm) /.  $\omega^k \rightarrow \omega$ ,
    DeleteCases[Times @@ ({A[1], B[1], P[1], Q[1]}^#) & /@ Tuples[Range[0, 2], 4], 1]]
```

```
Out[33]= 1 - 2 Z^3  $\omega$  + Z^6  $\omega$  - Z^4  $\omega$  A[1]  $\times$  B[1] + Z^3  $\omega$  A[1]  $\times$  P[1] + Z^3  $\omega$  B[1]  $\times$  Q[1] - Z^2 P[1]  $\times$  Q[1]
```

```
In[34]:= InvLSymm3a = SymmetricReduction[InvL[U, 3, a], {a[1], a[2], a[3]}, {A[1], P[1],  $\omega$ }]][1]
```

```
Out[34]= 1 - U^3  $\omega$  - U A[1] + U^2 P[1]
```

```
In[35]:= InvLSymm3b = SymmetricReduction[InvL[V, 3, b], {b[1], b[2], b[3]}, {B[1], Q[1],  $\omega$ }]][1]
```

```
Out[35]= 1 - V^3  $\omega$  - V B[1] + V^2 Q[1]
```

```

In[36]:= GL3inSym = Expand[FullSimplify[(Part1Symm * InvLSymm3a +
      Part2Symm * InvLSymm3b - Part3Symm * InvLSymm3a * InvLSymm3b)]] /. {ω^k_ -> ω};

In[37]:= (*Ramified Case*)

In[38]:= RamListMono3 = List@@(GL3inSym /. ω -> 0)
Out[38]=
{1, -U V A[1] × B[1], U2 V B[1] × P[1], -U Z B[1] × P[1], U V Z B[1]2 P[1], U V2 A[1] × Q[1], -V Z A[1] × Q[1],
  U V Z A[1]2 Q[1], -U2 V2 P[1] × Q[1], -Z2 P[1] × Q[1], -U2 V Z A[1] × P[1] × Q[1], U Z2 A[1] × P[1] × Q[1],
  -U V2 Z B[1] × P[1] × Q[1], V Z2 B[1] × P[1] × Q[1], -U V Z2 A[1] × B[1] × P[1] × Q[1], U2 V2 Z2 P[1]2 Q[1]2}

In[39]:= Length[RamListMono3]
Out[39]=
16

In[40]:= RamListexpo3 =
  Exponent[#, p] & /@ (List@@ RamListMono3 /. {A[1] -> p^Th, B[1] -> p^Th, P[1] -> p^(2 * Th),
    Q[1] -> p^(2 * Th), U -> p^(-1/2), V -> p^(-1/2), Z -> p^(-1)} /. Th -> 5/14) // N;

In[41]:= Total[RamListMono3[[#]] & /@ Flatten@Position[#, < 0 & /@ RamListexpo3, False]]
Out[41]=
1

In[42]:= (*Unramified Case*)

In[43]:= UnrListMono3 = List@@(Expand[GL3inSym /. ω -> 1]);

In[44]:= Length[UnrListMono3]
Out[44]=
158

In[45]:= UnrListexpo3 =
  Exponent[#, p] & /@ (List@@ UnrListMono3 /. {A[1] -> p^Th, B[1] -> p^Th, P[1] -> p^Th,
    Q[1] -> p^Th, U -> p^(-1/2), V -> p^(-1/2), Z -> p^(-1)} /. Th -> 5/14) // N;

In[46]:= (*The part of the arithmetic factor \mathfrak
  {A}_{\mathfrak{p}} that can't be bounded by Kim-Sarnak alone*)

In[47]:= Total[UnrListMono3[[#]] & /@ Flatten@Position[#, < -1 & /@ UnrListexpo3, False]]
Out[47]=
1 - U V A[1] × B[1] + U2 V B[1] × P[1] - U Z B[1] × P[1] + U V Z B[1]2 P[1] + U V2 A[1] × Q[1] - V Z A[1] × Q[1] + U V Z A[1]2 Q[1]

In[48]:= Length[Total[UnrListMono3[[#]] & /@ Flatten@Position[#, < -1 & /@ UnrListexpo3, False]]]
Out[48]=
8

```